

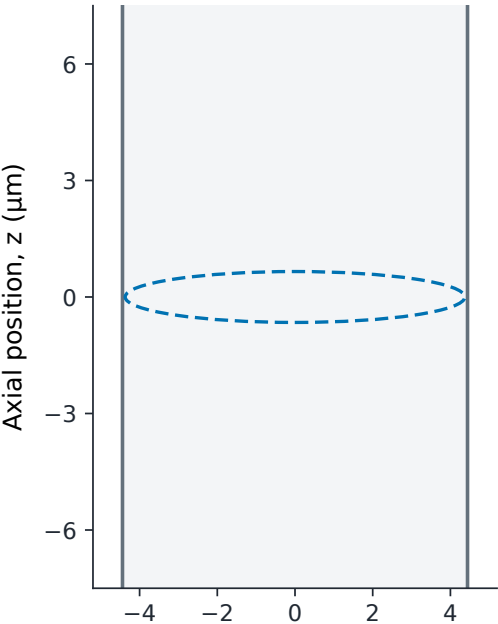
# Geometric feasibility of an RBC deformation surrogate

Same reference cell:  $V_0 = 52.439 \text{ fL}$ ;  $A_{\text{max}} = 127.079 \text{ }\mu\text{m}^2$

--- Original oblate geometry    — Selected capsule    — Tube wall

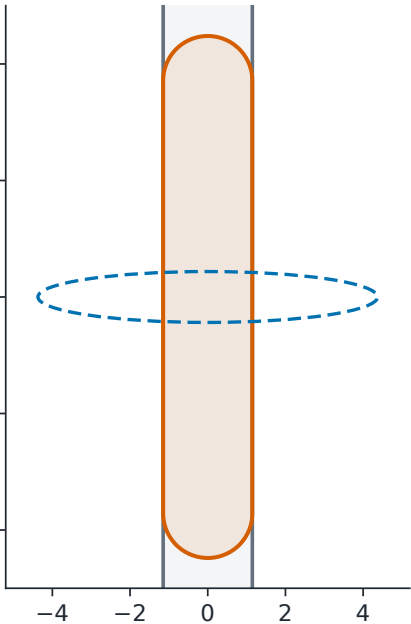
**(a) Free oblate**

Wall half-width  $h = 4.440 \text{ }\mu\text{m}$



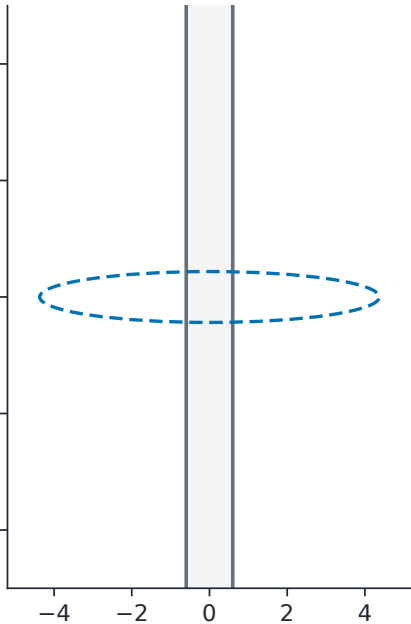
**(b) Feasible capsule**

Wall half-width  $h = 1.148 \text{ }\mu\text{m}$



**(c) Surrogate infeasible**

Wall half-width  $h = 0.599 \text{ }\mu\text{m}$



| Geometric quantity                     | (a) Free | (b) Feasible | (c) Infeasible |
|----------------------------------------|----------|--------------|----------------|
| Wall clearance ( $\mu\text{m}$ )       | 0.070    | 0.000        | —              |
| Capsule radius, $R$ ( $\mu\text{m}$ )  | —        | 1.148        | —              |
| Cylinder length, $L$ ( $\mu\text{m}$ ) | —        | 11.146       | —              |
| Capsule area / area budget             | —        | 0.763        | —              |

Geometric surrogate only; infeasibility does not imply physiological occlusion.